

IndiaOS MVP — PostgreSQL Database Schema

Status: Locked for MVP

Target: Supabase (PostgreSQL 14+)

Last Updated: June 2026

Revisions: 3 (schema refinements applied)

Executive Summary

16 tables organized into 7 functional tiers. **17 enums** define state and type values.

Row-level security (RLS) enforces citizen privacy. **Immutable records** (soft deletes, append-only audit trails, versioned snapshots) ensure governance memory survives elections and external pressure.

Core insight: Reports are private inputs; issues are public outputs. AI clustering discovers issues; citizens never create them directly.

ENUM DEFINITIONS (17 types)

1. citizen_status_enum

active | inactive | suspended

Citizen account status. inactive = soft deactivation (user can reactivate). suspended = admin-initiated restriction.

2. report_status_enum

submitted | verified | disputed | archived

Internal report lifecycle. Only verified reports feed confidence scoring.

3. verification_level_enum

phone_verified | aadhaar_verified | government_verified

Progressively higher assurance of citizen identity. Seeds Trust Score.

4. official_status_enum

unacknowledged | acknowledged | in_progress | completed | dismissed

[REFINED] Explicit enum instead of NULL-reliant.

Why: WHERE official_status = 'unacknowledged' is cleaner than WHERE official_status IS NULL for operational queries.

- unacknowledged — Government has not yet responded (initial state).

- acknowledged — Government confirms issue; response initiated.
- in_progress — Government reports active work.
- completed — Government claims resolution.
- dismissed — Government rejects as invalid.

Default on issue creation: unacknowledged

5. reality_status_enum

emerging | verified | trending | in_progress | partially_resolved | resolved | archived

Evidence-driven status independent of official claims. High confidence + citizen reports = trending. Evidence of fix = partially_resolved or resolved. No recent reports + time elapsed = archived.

6. responsibility_type_enum

primary | secondary | supporting

Relative accountability weight in issue_governance_entities.

7. evidence_source_type_enum

news_outlet | rti_portal | government_agency | court | academic | ngo | sensor | other

Seedable credibility categories. Seeds evidence_sources.credibility_score.

8. evidence_type_enum

news_article | rti_response | government_document | court_order | photo | video | audio
| sensor_data | other

Content classification for ERS (Evidence Reliability Score) weighting.

9. governance_level_enum

national | state | district | city | ward

Hierarchy for filtering and jurisdiction assignment.

10. entity_type_enum

government | department | agency | municipality | other

Authority type for abstract governance_entities model. Hardcoded on creation; no evolution per entity.

11. update_type_enum

tender_approved | budget_released | work_started | progress_update | milestone | completion | delay

Operational milestone types in issue_updates. Only these appear in public issue pages.

12. outcome_type_enum

funds_released | tender_approved | project_started | project_completed | milestone | policy_change | other

Measurable results in issue_outcomes.

13. lifecycle_event_type_enum

created | verified | trending | in_progress | partially_resolved | resolved | reopened | archived

State transitions in issue_lifecycle_events. Immutable append-only record.

14. trigger_type_enum

ai_clustering | momentum_threshold | government_action | evidence_update | rgs_reopen | citizen_flag | admin_override

Why did a lifecycle event fire?

15. claimed_status_enum

acknowledged | in_progress | completed | dismissed

Subset of official_status_enum for resolution_claims (government assertions, not issue state).

16. message_type_enum

text | photo | video | audio | document

WhatsApp message content type for bot_conversations.

17. message_status_enum

sent | delivered | read | failed

WhatsApp delivery status; seeds audit trail.

TABLE SPECIFICATION (16 tables × 7 tiers)

TIER 1: CORE CITIZENS & REPORTS

citizens (Immutable citizen registry)

Field	Type	Constraints	Notes
id	UUID	PK	Immutable. Never hard-deleted.
phone_hash	VARCHAR(128)	UNIQUE NOT NULL	SHA256(phone_number + salt). Phone never stored plaintext.
phone_verified_at	TIMESTAMPTZ		NULL until SMS OTP confirmed.
verification_level	verification_level_enum	NOT NULL DEFAULT 'phone_verified'	Seeds Trust Score.
status	citizen_status_enum	NOT NULL DEFAULT 'active'	Soft deactivation: UPDATE status='inactive', never DELETE.
status_changed_at	TIMESTAMPTZ		Track when status changed.
created_at	TIMESTAMPTZ	NOT NULL	Immutable.
last_activity_at	TIMESTAMPTZ		Updated on every report/message.
report_count	INT	DEFAULT 0	Denormalized for ranking; recomputed nightly.

Constraints:

- TRIGGER — Prevent DELETE; log attempts.
- FK citizens.id in reports: ON DELETE RESTRICT (cannot delete citizen with reports).

RLS:

- Citizen row: SELECT only own row. No UPDATE/DELETE from citizen role.
 - Official/admin: No access to citizen personal data (phone hash, verification, status).
-

reports (Immutable private submissions)

Field	Type	Constraints	Notes
id	UUID	PK	Never regenerated.
citizen_id	UUID	FK NOT NULL	ON DELETE RESTRICT.
issue_id	UUID	FK NULLABLE	ON DELETE SET NULL. AI assigns post-clustering. [Future: migrate to issue_reports junction table in Phase 2+ for multi-issue support.]
category_id	INT	FK NOT NULL	ON DELETE RESTRICT.
location_id	INT	FK NOT NULL	ON DELETE RESTRICT.
title	VARCHAR(255)	NOT NULL	User-provided headline.
description	TEXT	NOT NULL	User-provided details.
latitude	NUMERIC(9,6)	NOT NULL	CHECK: -90 to +90
longitude	NUMERIC(9,6)	NOT NULL	CHECK: -180 to +180
status	report_status_enum	NOT NULL DEFAULT 'submitted'	Only 'verified' feeds scoring.
is_duplicate	BOOLEAN	DEFAULT FALSE	AI clustering flag.
ai_clustering_decision	JSONB		Serialized clustering metadata from Governance Intelligence Engine.
created_at	TIMESTAMPTZ	NOT NULL	Immutable.
verified_at	TIMESTAMPTZ		Set when status='verified'.
archived	BOOLEAN	DEFAULT FALSE	Soft archive flag.

Constraints:

- TRIGGER — Prevent hard DELETE; log attempts.
- No photo_urls JSON array (moved to reports_media for cleaner schema).

RLS:

- Citizen role: SELECT own reports only. No INSERT (app handles via API route with session validation).
- Official/admin: Cannot see raw reports directly. See only aggregated issues table (public).

reports_media (Report photo/video/document metadata)

Field	Type	Constraints	Notes
id	UUID	PK	
report_id	UUID	FK NOT NULL	ON DELETE CASCADE.
file_path	VARCHAR(512)	UNIQUE NOT NULL	Format: reports_media/{report_uuid}/{media_type}/{filename}. Points to Supabase Storage, not embedded.
media_type	message_type_enum	NOT NULL	photo, video, audio, document.
file_size_bytes	BIGINT	NOT NULL	For quota enforcement.
original_filename	VARCHAR(255)		User-uploaded name (for display).
mime_type	VARCHAR(64)		Content-type.
uploaded_at	TIMESTAMPTZ	NOT NULL	
archived	BOOLEAN	DEFAULT FALSE	Soft archive flag.

Constraints:

- TRIGGER — Set archived=TRUE if parent report is archived.

RLS:

- Citizen: SELECT own media. Upload path: reports_media/{citizen_uuid}/{report_uuid}/{media_type}/{filename}.
- Official/admin: No direct media access (privacy).

bot_conversations (WhatsApp audit trail)

Field	Type	Constraints	Notes
id	UUID	PK	
citizen_id	UUID	FK NOT NULL	ON DELETE CASCADE.
message_id	VARCHAR(64)	UNIQUE NOT NULL	Twilio message SID; prevents duplicate inserts.
direction	ENUM ('inbound', 'outbound')	NOT NULL	Incoming citizen message or bot response.
message_text	TEXT		Full text; encrypted at rest in production.
message_type	message_type_enum	NOT NULL	text, photo, video, audio, document.
timestamp	TIMESTAMPTZ	NOT NULL	When Twilio created/delivered message.
status	message_status_enum	NOT NULL	sent, delivered, read, failed.
associated_report_id	UUID	FK NULLABLE	ON DELETE SET NULL. If message generated a report, link it.
created_at	TIMESTAMPTZ	NOT NULL	When record inserted ($\approx$ timestamp).

Constraints:

- Webhook handler inserts with ON CONFLICT (message_id) DO NOTHING for idempotency.

RLS:

- Citizen: SELECT own conversations. Admin: SELECT all for audit.

TIER 2: ISSUE MANAGEMENT

issues (Public clusters of related reports)

Field	Type	Constraints	Notes
id	UUID	PK	Created by AI service only.
title	VARCHAR(255)	NOT NULL	Cluster headline (AI-generated or human-edited).
description	TEXT	NOT NULL	Synthesis of related reports.
category_id	INT	FK NOT NULL	ON DELETE RESTRICT.
location_id	INT	FK NOT NULL	ON DELETE RESTRICT.
latitude	NUMERIC(9,6)	CHECK -90:90	Geographic center of cluster.
longitude	NUMERIC(9,6)	CHECK -180:180	
official_status	official_status_enum	NOT NULL DEFAULT 'unacknowledged'	[REFINED] Explicit enum, no NULL. Government response state.
reality_status	reality_status_enum	NOT NULL DEFAULT 'emerging'	Evidence-driven state. Independent of official claims.
confidence_score	NUMERIC(3,2)	CHECK 0:1	0.00–1.00. CS from Governance Intelligence Engine, cached here.

Field	Type	Constraints	Notes
momentum_score	NUMERIC(3,2)	CHECK 0:1	0.00–1.00. Report velocity + citizen engagement.
report_count	INT	DEFAULT 0	Denormalized. Recomputed on daily batch.
evidence_count	INT	DEFAULT 0	Denormalized. Count of linked evidence records.
reports_24h	INT	DEFAULT 0	Reports in last 24h.
reports_7d	INT	DEFAULT 0	Reports in last 7d.
reports_30d	INT	DEFAULT 0	Reports in last 30d.
created_at	TIMESTAMPTZ	NOT NULL	When AI clustering created issue.
updated_at	TIMESTAMPTZ	NOT NULL	Last merge, split, or evidence/claim update.
last_activity_at	TIMESTAMPTZ	NOT NULL	Last report/evidence/claim linked.
is_active	BOOLEAN	DEFAULT TRUE	FALSE if archived.

Constraints:

- No hard DELETE. Set is_active=FALSE for archival.
- TRIGGER — If official_status='completed' and reality_status NOT IN ('resolved', 'partially_resolved'), compute Reality Gap Score (RGS). If RGS > 0.60, reopen: set official_status='in_progress', fire lifecycle event event_type='reopened'.

RLS:

- Fully public SELECT (citizens, officials, anonymous). Aggregates only; no raw data leakage.
- INSERT/UPDATE: system role (AI service) only.
- DELETE: Restricted; use soft deactivation.

issue_updates (Operational progress tracking)

Field	Type	Constraints	Notes
id	UUID	PK	
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE.
governance_entity_id	UUID	FK NULLABLE	ON DELETE SET NULL. Which agency posted this update?
update_type	update_type_enum	NOT NULL	tender_approved, budget_released, work_started, progress_update, milestone, completion, delay.
title	VARCHAR(255)	NOT NULL	Headline: e.g., "Tender issued for pothole repair"
description	TEXT	NOT NULL	Details.
progress_percent	SMALLINT	CHECK 0:100	0–100. For work_started, progress_update, milestone, completion.
evidence_id	UUID	FK NULLABLE	ON DELETE SET NULL. Supporting evidence (news, RTI, government document).
created_at	TIMESTAMPTZ	NOT NULL	When update posted.
archived	BOOLEAN	DEFAULT FALSE	Soft archive for old/retracted updates.

Constraints:

- TRIGGER — On INSERT, update issues.updated_at, issues.last_activity_at.

RLS:

- Fully public SELECT.
- INSERT: governance_entity_id must match authenticated entity role (via JWT), or system role.

TIER 3: ORGANIZATION & HIERARCHY

categories (Hierarchical issue taxonomy)

Field	Type	Constraints	Notes
id	SERIAL	PK	
name	VARCHAR(128)	UNIQUE NOT NULL	"Roads", "Water", "Potholes", etc.
parent_category_id	INT	FK NULLABLE	ON DELETE SET NULL. Self-referential hierarchy.
description	TEXT		What issues belong here.
sla_days_benchmark	SMALLINT		Target days to resolve. For GHS computation.
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- FK categories.parent_category_id → categories.id: ON DELETE SET NULL

RLS:

- Fully public SELECT. No UPDATE/DELETE from citizen/official roles.

locations (Geographic hierarchy)

Field	Type	Constraints	Notes
id	SERIAL	PK	
name	VARCHAR(128)	NOT NULL	"Maharashtra", "Pune", "Ward 42", etc.
location_type	VARCHAR(64)		"state", "district", "city", "ward", "neighborhood". Free text for flexibility.
parent_location_id	INT	FK NULLABLE	ON DELETE SET NULL. Self-referential hierarchy.
latitude	NUMERIC(9,6)		Center of region (for map display).

Field	Type	Constraints	Notes
longitude	NUMERIC(9,6)		
population	INT	NULLABLE	For per-capita metrics (GHS).
created_at	TIMESTAMPTZ	NOT NULL	
Constraints:			
CHECK: latitude -90 to +90, longitude -180 to +180.			
RLS:			
Fully public SELECT.			

TIER 4: GOVERNANCE & ACCOUNTABILITY

governance_entities (Abstract authority model)

Field	Type	Constraints	Notes
id	UUID	PK	
name	VARCHAR(255)	NOT NULL	"Ministry of Roads", "Pune Municipal Corporation", "District Water Board".
entity_type	entity_type_enum	NOT NULL	government, department, agency, municipality, other.
governance_level	governance_level_enum	NOT NULL	national, state, district, city, ward. For jurisdiction filtering.
location_id	INT	FK NOT NULL	ON DELETE RESTRICT. Where entity operates.
parent_entity_id	UUID	FK NULLABLE	ON DELETE SET NULL. Hierarchical: Ministry → Department → Bureau.
contact_email	VARCHAR(255)		Official email for notifications.
contact_phone	VARCHAR(20)		Phone for escalations.
website_url	VARCHAR(255)		Public website.

Field	Type	Constraints	Notes
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- No self-delete if has children; parent_entity_id SET NULL on parent delete.

RLS:

- Fully public SELECT. No UPDATE/DELETE from official role.

governance_terms (Election cycle tracking)

Field	Type	Constraints	Notes
id	UUID	PK	
governance_entity_id	UUID	FK NOT NULL	ON DELETE CASCADE.
term_start_date	DATE	NOT NULL	When this term began.
term_end_date	DATE	NULLABLE	When it ends (NULL = ongoing).
ruling_party_name	VARCHAR(128)		e.g., "Indian National Congress".
chief_executive_name	VARCHAR(255)		e.g., "Arvind Kejriwal" (Chief Minister).
notes	TEXT		Election results, coalition notes, etc.
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- **TRIGGER — IMMUTABLE.** BEFORE UPDATE ON governance_terms FOR EACH ROW RAISE EXCEPTION 'governance_terms is immutable'. Governance Memory is permanent. Elections transition by inserting new term, not updating old one.

RLS:

- Fully public SELECT.

issue_governance_entities (M:N — Issues ↔ Responsible Authorities)

Field	Type	Constraints	Notes
id	UUID	PK	
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE.
governance_entity_id	UUID	FK NOT NULL	ON DELETE CASCADE.
responsibility_type	responsibility_type_enum	NOT NULL	primary, secondary, supporting. Primary entity is first to respond.
assigned_at	TIMESTAMPTZ	NOT NULL	When entity became responsible.
acknowledgment_date	DATE	NULLABLE	When entity first acknowledged issue.
acknowledgment_text	TEXT	NULLABLE	Government's acknowledgment statement.
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- UNIQUE(issue_id, governance_entity_id) — One entity per issue per responsibility tier. (Can have primary + secondary + supporting simultaneously.)

RLS:

- Fully public SELECT.
- INSERT/UPDATE: Governance entity or system role only.

TIER 5: EVIDENCE & VERIFICATION

evidence_sources (Curated source registry)

Field	Type	Constraints	Notes
id	SERIAL	PK	

Field	Type	Constraints	Notes
source_name	VARCHAR(255)	UNIQUE NOT NULL	"The Hindu", "Press Information Bureau", "Lok Sabha", "IndiaStack RTI API", "Mapbox".
source_type	evidence_source_type_enum	NOT NULL	news_outlet, rti_portal, government_agency, court, academic, ngo, sensor, other.
credibility_score	NUMERIC(3,2)	CHECK 0:1	[SEED VALUE] Starting credibility (news = 0.70, government = 0.85, sensor = 0.65). Refined in Phase 2+ via configurable trusted_sources table.
url_pattern	VARCHAR(512)	NULLABLE	Regex for auto-detecting articles from this source.
is_verified	BOOLEAN	DEFAULT FALSE	Source independently validated (WHOIS, DNS, legal registration).
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- Seed data at deployment. Hardcoded values for MVP.

RLS:

- Fully public SELECT.

evidence (Evidence items linked to issues)

[Formerly evidence_issues]

Field	Type	Constraints	Notes
id	UUID	PK	

Field	Type	Constraints	Notes
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE. Which issue does this support?
evidence_source_id	INT	FK NOT NULL	ON DELETE RESTRICT. From which source?
evidence_url	VARCHAR(512)	NOT NULL	Link to original (news article, RTI, court document, sensor data).
evidence_title	VARCHAR(255)	NOT NULL	Headline/citation.
evidence_type	evidence_type_enum	NOT NULL	news_article, rti_response, government_document, court_order, photo, video, audio, sensor_data, other.
ers_score	NUMERIC(3,2)	CHECK 0:1	Evidence Reliability Score (0.00–1.00). Computed by Governance Intelligence Engine. = source_credibility × freshness × semantic_match.
extracted_text	TEXT	NULLABLE	OCR or plain-text extraction (for searching).
semantic_match_score	NUMERIC(3,2)	CHECK 0:1	How strongly does evidence match issue description? 0–1.
published_at	DATE	NULLABLE	Original publication date.
source_name	VARCHAR(255)		Cached source name (denormalized for query speed).
author	VARCHAR(255)	NULLABLE	Article author, RTI requestor, court case parties.

Field	Type	Constraints	Notes
is_disputed	BOOLEAN	DEFAULT FALSE	Counter-evidence exists. Marks this evidence as contested.
created_at	TIMESTAMPTZ	NOT NULL	When record inserted.

Constraints:

- TRIGGER — On INSERT, update issues.evidence_count, issues.updated_at, issues.last_activity_at.

RLS:

- Fully public SELECT.
- INSERT: system role (AI service crawling + semantic matching) only.

resolution_claims (Government assertions vs. reality)

Field	Type	Constraints	Notes
id	UUID	PK	
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE. Which issue?
governance_entity_id	UUID	FK NOT NULL	ON DELETE CASCADE. Which entity claimed resolution?
claim_number	SMALLINT	NOT NULL	Sequence for same issue+entity (1st claim, 2nd claim, etc.).
claimed_status	claimed_status_enum	NOT NULL	What status is entity claiming? (acknowledged, in_progress, completed, dismissed).
claim_date	DATE	NOT NULL	When entity made this claim.

Field	Type	Constraints	Notes
claim_evidence_url	VARCHAR(512)	NULLABLE	Government's supporting link (e.g., tender award, budget release, photo).
claim_evidence_id	UUID	FK NULLABLE	ON DELETE SET NULL. Link to supporting evidence record (if in our DB).
reality_status_at_claim	reality_status_enum	NOT NULL	IMMUTABLE SNAPSHOT. What was the reality status when claim was made? Never updated.
current_reality_status	reality_status_enum	NOT NULL	What is reality status NOW? Recomputed nightly.
is_valid	BOOLEAN	DEFAULT NULL	NULL = monitoring. TRUE = claims match evidence. FALSE = contradicted by reports/evidence. Recomputed nightly.
reason_if_invalid	TEXT	NULLABLE	Why is claim invalid? "Citizen reports still arriving", "RTI contradicts claim", etc.
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- UNIQUE(issue_id, governance_entity_id, claim_date) — One claim per entity per issue per date.
- **IMMUTABLE:** reality_status_at_claim never changes post-insertion.

RLS:

- Fully public SELECT (transparency: show both claims and reality).
 - INSERT: governance_entity_id or system role only.
-

TIER 6: OUTCOMES & ACCOUNTABILITY AUDIT

issue_outcomes (Measurable results)

Field	Type	Constraints	Notes
id	UUID	PK	
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE.
governance_entity_id	UUID	FK NOT NULL	ON DELETE CASCADE. Which entity delivered the outcome?
outcome_type	outcome_type_enum	NOT NULL	funds_released, tender_approved, project_started, project_completed, milestone, policy_change, other.
amount_inr	BIGINT	NULLABLE	Rupees released (for fund outcomes).
outcome_date	DATE	NOT NULL	When outcome achieved.
outcome_description	TEXT	NOT NULL	Details: tender ID, project name, amount, beneficiaries.
confirming_evidence_id	UUID	FK NULLABLE	ON DELETE SET NULL. Evidence proving outcome (news, RTI, government press release).
confirmed_date	DATE	NULLABLE	When independently verified.
beneficiaries_count	INT	NULLABLE	How many people affected? For per-capita metrics.
created_at	TIMESTAMPTZ	NOT NULL	

Constraints:

- TRIGGER — On INSERT, update issues.updated_at, issues.last_activity_at.

RLS:

- Fully public SELECT.

issue_lifecycle_events (Immutable audit trail)

Field	Type	Constraints	Notes
id	UUID	PK	
issue_id	UUID	FK NOT NULL	ON DELETE CASCADE. Which issue?
event_type	lifecycle_event_type_enum	NOT NULL	created, verified, trending, in_progress, partially_resolved, resolved, reopened, archived.
triggered_by	trigger_type_enum	NOT NULL	ai_clustering, momentum_threshold, government_action, evidence_update, rgs_reopen, citizen_flag, admin_override.
triggered_by_entity_id	UUID	FK NULLABLE	ON DELETE SET NULL. If government_action or admin_override, which entity?
confidence_score_at_event	NUMERIC(3,2)		Snapshot of CS at this moment.
momentum_score_at_event	NUMERIC(3,2)		Snapshot of momentum.

Field	Type	Constraints	Notes
report_count_at_event	INT		How many reports when event fired?
reason	TEXT	NOT NULL	Why did event fire? "CS crossed 0.85", "Momentum fell below 0.30", "Government acknowledged", etc.
evidence_snapshot	JSONB	NULLABLE	Serialized evidence metadata at event time (sources, ERS scores).
created_at	TIMESTAMPTZ	NOT NULL	IMMUTABLE.

Constraints:

- **TRIGGER — APPEND-ONLY.** BEFORE UPDATE ON issue_lifecycle_events FOR EACH ROW RAISE EXCEPTION 'Lifecycle events are immutable'. Governance memory is permanent.
- **TRIGGER — BEFORE DELETE** ON issue_lifecycle_events FOR EACH ROW RAISE EXCEPTION 'Lifecycle events cannot be deleted'.

RLS:

- Fully public SELECT (transparency).
- INSERT: system role only.

ROW-LEVEL SECURITY (RLS)

Roles

citizen — Authenticated IndiaOS user (phone-verified via WhatsApp).

official — Government employee (entity_id verified via JWT).

admin — IndiaOS staff (full access for operations, moderation, config).

system — Privileged backend service account (AI service + scheduled jobs).

anon — Unauthenticated browser user (public reads only).

AI Service Role (NEW)

Purpose: Privileged backend account for the Governance Intelligence Engine.

Identity: Service account authenticated via API key (server-side, never exposed in frontend).

Permissions:

- **READ** — All reports (to cluster), all issues, all evidence, all resolution claims, all categories, all locations, all governance entities.
- **INSERT** — issues (create clusters), issue_updates (post operational progress), evidence (from news/RTI crawlers), issue_lifecycle_events (state transitions), issue_governance_entities (assign responsibility).
- **UPDATE** — issues (merge, split, update scores, reopen on RGS). Resolution claims: compute is_valid, recompute current_reality_status nightly.
- **DELETE** — None. Soft-deletes only.

No audit bypass: Every AI action is logged in issue_lifecycle_events with triggered_by='ai_clustering' or similar. Citizens see which decisions were AI-driven.

Policy Summary

Table	Citizen	Official	Admin	System	Anon
citizens	Own row SELECT	X	SELECT all	X	X
reports	Own SELECT	X	SELECT all	READ all	X
reports_media	Own SELECT	X	SELECT all	READ all	X
bot_conversations	Own SELECT	X	SELECT all	INSERT/UPDATE	X
issues	SELECT all	SELECT all	SELECT all	INSERT/UPDATE	SELECT all
issue_updates	SELECT all	SELECT/INSERT (own entity)	SELECT all	INSERT	SELECT all
categories	SELECT all	SELECT all	SELECT all	X	SELECT all

Table	Citizen	Official	Admin	System	Anon
locations	SELECT all	SELECT all	SELECT all	X	SELECT all
governance_entities	SELECT all	SELECT all	SELECT all	X	SELECT all
governance_terms	SELECT all	SELECT all	SELECT all	X	SELECT all
issue_governance_entities	SELECT all	SELECT/INSERT (own entity)	SELECT all	INSERT/UPDATE	SELECT all
evidence_sources	SELECT all	SELECT all	SELECT all	X	SELECT all
evidence	SELECT all	SELECT all	SELECT all	INSERT	SELECT all
resolution_claims	SELECT all	SELECT/INSERT (own entity)	SELECT all	INSERT/UPDATE	SELECT all
issue_outcomes	SELECT all	SELECT/INSERT (own entity)	SELECT all	INSERT	SELECT all
issue_lifecycle_events	SELECT all	SELECT all	SELECT all	INSERT	SELECT all

INDEXES (12 core + Phase 2+)

MVP Core (Query Performance)

```
CREATE INDEX idx_reports_location_created ON reports(location_id, created_at DESC)
WHERE archived=false;
```

```
CREATE INDEX idx_reports_issue ON reports(issue_id);
```

```
CREATE INDEX idx_issues_confidence_location ON issues(confidence_score DESC,
location_id) WHERE is_active=true;
```

```
CREATE INDEX idx_issues_momentum ON issues(momentum_score DESC) WHERE
is_active=true;
```

```
CREATE INDEX idx_evidence_issue ON evidence(issue_id);
```

```
CREATE INDEX idx_lifecycle_issue_date ON issue_lifecycle_events(issue_id, created_at  
DESC);
```

```
CREATE INDEX idx_issue_gov_entity_entity ON  
issue_governance_entities(governance_entity_id);
```

```
CREATE INDEX idx_resolution_claims_entity ON  
resolution_claims(governance_entity_id, is_valid);
```

```
CREATE INDEX idx_citizens_active ON citizens(status) WHERE status='active';
```

```
CREATE INDEX idx_issue_updates_issue ON issue_updates(issue_id, created_at  
DESC);
```

```
CREATE INDEX idx_governance_entities_level ON  
governance_entities(governance_level);
```

Phase 2+ (Full-Text + Aggregation)

```
CREATE INDEX idx_issues_search ON issues USING GIN(to_tsvector('english',  
description));
```

```
CREATE MATERIALIZED VIEW issue_rankings AS
```

```
SELECT issues.id, issues.title, issues.location_id,
```

```
       issues.confidence_score, issues.momentum_score, issues.report_count,
```

```
       (issues.report_count * CASE WHEN issues.confidence_score > 0.80 THEN 1.5 ELSE  
1.0 END) as public_interest_score
```

```
FROM issues WHERE is_active=true
```

```
ORDER BY public_interest_score DESC;
```

```
CREATE INDEX idx_issue_rankings ON issue_rankings(location_id,  
public_interest_score DESC);
```

```
CREATE MATERIALIZED VIEW location_health_scores AS
```

```
SELECT locations.id,
```

```
       ROUND(AVG(CASE WHEN issues.reality_status IN ('resolved','partially_resolved')  
THEN 1.0 ELSE 0.0 END)::numeric, 2) as resolution_rate,
```

```
       COUNT(DISTINCT issues.id) as active_issue_count,
```

```
       AVG(issues.confidence_score) as avg_confidence
```



```
FROM locations LEFT JOIN issues ON issues.location_id = locations.id WHERE  
issues.is_active=true
```

```
GROUP BY locations.id;
```

DATA FLOW & OPERATIONAL SEMANTICS

Report → Issue Lifecycle

Citizen submits report via WhatsApp

↓ INSERT reports (citizen_id, category_id, location_id, status='submitted')

Report stored privately

↓ Bot extracts media → INSERT reports_media

↓ AI service reads unlinked reports

AI clusters similar reports

↓ Issue exists? UPDATE reports.issue_id = ?

↓ Issue doesn't exist? INSERT issues (title, description, confidence_score)

Report linked to issue

↓ TRIGGER: UPDATE issues.report_count, issues.updated_at, issues.last_activity_at

Recalculate confidence/momentum

↓ INSERT issue_lifecycle_events (event_type='verified', triggered_by='ai_clustering',
confidence_score_at_event=0.75)

Public issue page updated (aggregates only; report remains private)

Government Response Flow

Government acknowledges issue

↓ INSERT issue_governance_entities (responsibility_type='primary',
acknowledgment_date=TODAY)

↓ TRIGGER: UPDATE issues.official_status = 'acknowledged'

Issue officially recognized

Government posts progress

↓ INSERT issue_updates (update_type='tender_approved', progress_percent=NULL)

Citizens see milestones

Government claims completion

↓ INSERT resolution_claims (claimed_status='completed',
reality_status_at_claim='in_progress')

↓ TRIGGER: SET official_status='completed', start 30-day RGS monitoring

RGS monitor fires nightly

↓ Are new reports arriving? RGS score ↑

↓ If RGS > 0.60: UPDATE issues.official_status='in_progress', INSERT
issue_lifecycle_events (event_type='reopened', triggered_by='rgs_reopen')

Reality Gap visible: government claim contradicted by citizen evidence

DEFERRED TO PHASE 2+ (Documented)

Schema Additions

- **issue_reports (M:N junction)** — Replace reports.issue_id for multi-clustering support. Current design (1 report → 1 issue) sufficient for MVP. Post-clustering/splitting may require a report to belong to multiple issues. Migration path documented but deferred.
- **issue_occurrences** — Track recurrence: "pothole refilled 3 times in 18 months". Seeds governance memory patterns.
- **issue_lineages** — Merge/split history: "Issue #42 was split into #43, #44. Issues #100 and #101 merged into #120."
- **evidence_disputes** — Counter-evidence challenges. Current design: is_disputed flag + separate contradicting evidence records. Explicit dispute table deferred.
- **issue_followers** — Notification subscriptions. Citizens opt-in to updates on issues they care about.
- **trusted_sources (configurable)** — Override hardcoded evidence_sources.credibility_score. Admins tune source weights per category/region.
- **scoring_weights** — Versioned formula coefficients for CS, AS, RGS, GHS. Phase 2+ parametric tuning.

- **category_config** — SLA benchmarks per category per region. Current design: single sla_days_benchmark per category. Phase 2+ allows locality override.

Implementation Additions

- **Full-text search** via Postgres GIN indexes + Elasticsearch (Phase 2+).
 - **Partitioning** — reports by month, issue_lifecycle_events by year for sub-second query performance at scale.
 - **Archive strategy** — Move reports/media to cold storage (S3 glacier) after 7 years; keep DB records + metadata.
-

SCALABILITY & CAPACITY

Year 1 estimates:

- 50,000 active citizens
- 10,000 reports/day
- 10,000 public issues (clusters)
- 5-year archival horizon

Projected row counts:

- reports: ~3.65M rows (10k/day × 365 days × 1 year stored; older rows soft-archived)
- issues: ~10k rows
- evidence: ~200k rows
- issue_lifecycle_events: ~50k rows
- Total: ~5.3M rows

Storage:

- PostgreSQL: ~3.7 GB
- Supabase Storage (media): ~125 GB (assuming avg 12.5MB per report with photos)

Connection pooling:

- PgBouncer transaction mode: max_client_conn=1000

Query latency targets (p99):

- Public issue list: <200ms
 - Citizen report form: <100ms
 - Admin dashboard: <500ms
-

SECURITY NOTES

1. **Citizen phone never stored plaintext.** Always SHA256(phone + salt).
 2. **Message encryption at rest.** Twilio messages encrypted via pgcrypto or AWS KMS.
 3. **RLS enforced at database layer.** Not application-layer security.
 4. **Immutable records prevent tampering.** Triggers and soft deletes enforce data integrity.
 5. **Audit trail permanent.** issue_lifecycle_events and bot_conversations are append-only forever.
 6. **No bulk exports without approval.** Official/citizen roles cannot bulk-SELECT reports or media.
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